



JABchem



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Past Papers Advanced Higher Chemistry

2022 Marking Scheme

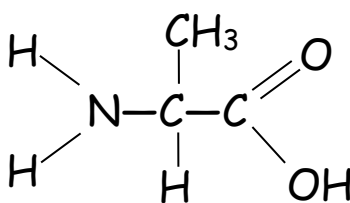
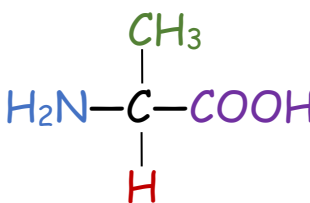
Grade Awarded	Mark Required		% candidates achieving grade
	(/120)	%	
A	77+	64.1%	32.6%
B	62+	51.7%	28.7%
C	48+	40%	20.7%
D	33+	27.5%	12.7%
No award	<33	<27.5%	5.3%

Section:	Multiple Choice	Extended Answer	Project
Average Mark:	15.7 /25	50.4 /95	No Project in 2022

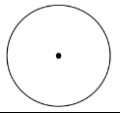
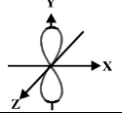
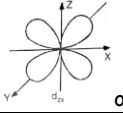
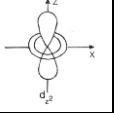
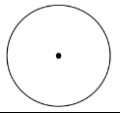
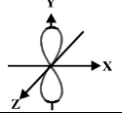
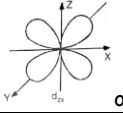
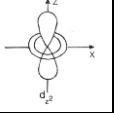
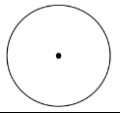
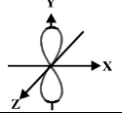
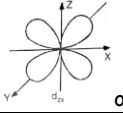
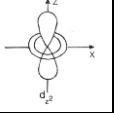
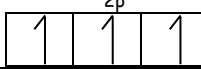
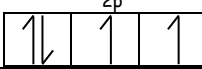
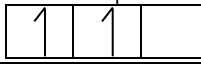
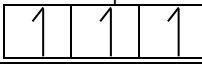
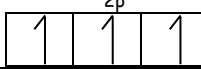
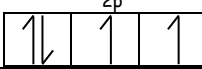
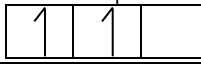
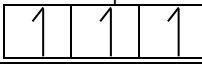
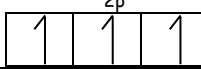
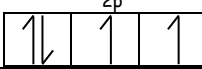
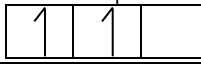
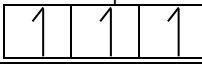
2022 Adv Higher Chemistry Marking Scheme

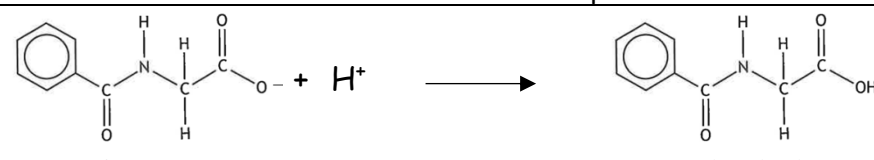
MC Qu	Answer	% Pupils Correct	Reasoning									
1	B	66	☒ A Each element does have a characteristic spectrum ☒ B Heat energy is used to promote electrons in emission spectroscopy ☒ C Lines are formed by electrons releasing energy as they drop down energy levels ☒ D More electrons dropping down levels leads to more intense lines formed									
2	D	60	Cr atom electronic configuration: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$ Cr ⁺ ion electronic configuration: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$									
3	D	61	Octahedral complex ion means M ²⁺ ion makes 6 dative covalent bonds with ligands. • L⁻ is a bidentate ligand making 2 dative covalent bonds with M²⁺ • M²⁺ pairs up with 3 L⁻ bidentate ligands • Formula of complex ion = [ML₃]⁻									
4	A	83	☒ A homogeneous catalysts are in the same state as the reactants ☒ B Adsorption of reactive molecules must take place in heterogeneous catalysis ☒ C A catalyst will reduce the activation energy for a reaction pathway ☒ D catalysts are often transition metals due to unpaired d electrons in mechanism									
5	D	41	☒ A Cl ⁻ is reduced to Cl ⁻ and I ⁻ is oxidised to I ₂ but are different substance ☒ B IO ₃ ⁻ is reduced to I ₂ and I ⁻ is oxidised to I ₂ but are different substances ☒ C Cl ₂ is reduced to Cl ⁻ and I ₂ is oxidised to IO ₃ ⁻ but are different substances ☒ D Cl ₂ is reduced to Cl ⁻ and at the same time Cl ₂ is oxidised to ClO ⁻ <table><tr><td>Reaction</td><td>Reduction reaction</td><td>Oxidation Reaction</td></tr><tr><td>Equation</td><td>Cl₂ + 2e⁻ → 2Cl⁻</td><td>Cl₂ + 2H₂O → 2ClO⁻ + 4H⁺ + 2e⁻</td></tr><tr><td>Oxidation State</td><td>0 -1</td><td>0 +1</td></tr></table>	Reaction	Reduction reaction	Oxidation Reaction	Equation	Cl ₂ + 2e ⁻ → 2Cl ⁻	Cl ₂ + 2H ₂ O → 2ClO ⁻ + 4H ⁺ + 2e ⁻	Oxidation State	0 -1	0 +1
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Oxidation State	0 -1	0 +1										
6	C	45	☒ A pH = 7.00 is true at 25°C (298K) ☒ B The ionic product K _w =[H ₃ O ⁺][OH ⁻] = 1.01x10 ⁻¹⁴ is only true at 25°C (298K) ☒ C [H ₃ O ⁺] = [OH ⁻] is true for any neutral solution at any temperature ☒ D [H ₃ O ⁺] = 1.00x10 ⁻⁷ at 25°C (298K)									
7	A	54	☒ A [SO ₃] will increase with an increase in pressure but value of K is unchanged ☒ B Changes to pressure do not change the value of the equilibrium constant K ☒ C Increase in pressure will favour pressure-reducing forward reaction ∴ ↑ [SO ₃] ☒ D Increase in pressure will favour pressure-reducing forward reaction ∴ ↑ [SO ₃]									
8	D	62	enthalpy of formation is the energy change for the formation of 1mol of a substance from its elements in their natural state. ☒ A Reactants are ions and not elements in their natural state ☒ B Reactants are ions and not elements in their natural state ☒ C Cl _(g) is not the natural state of chlorine Cl _{2(g)} ☒ D Reactants are elements in their natural state and 1 mole of product is formed									
9	B	83	Rate = k[A] ² [B] ∴ [A] ² means A is 2nd order (caused by having two particles of A in rate determining step) ∴ [B] ¹ means B is 1st order (caused by having one particles of B in rate determining step) ☒ A Doubling [A] leads to a quadrupling of reaction rate as A is 2 nd order ☒ B Doubling [B] leads to a doubling of reaction rate as B is 1 st order ☒ C Doubling [B] leads to a doubling of reaction rate as B is 1 st order ☒ D Doubling [A] and [B] would lead to a combined quadrupling of reaction rate.									
10	D	77	☒ A The order of each reactant is not decided by stoichiometric coefficients ☒ B The order of a reaction is decided by the slow rate determining step ☒ C The speed of reaction is determined by the slowest step - rate determining step ☒ D The individual orders are determined by experimentally altering concentrations and looking to see the effect on the reaction rate.									

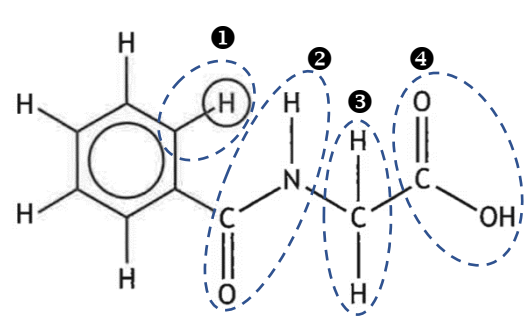
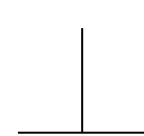
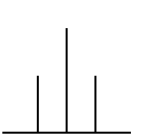
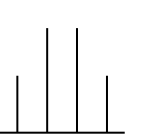
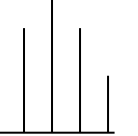
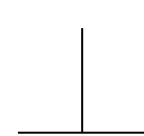
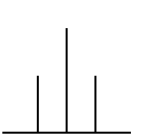
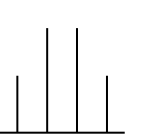
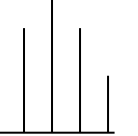
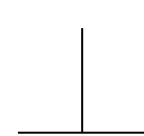
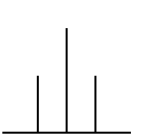
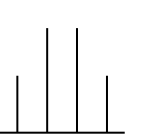
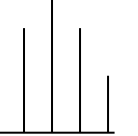
11	C	65	☒A there are no changes of state in this graph (indicated by vertical sections) ☒B there are no changes of state in this graph (indicated by vertical sections) ☒C Evaporation (2nd vertical section) gives a bigger increase in entropy than melting ☒D Evaporation (2nd vertical section) gives a bigger increase in entropy than melting																									
12	C	54	☒A diagram shows sp³ hybridisation as one 2s and three 2p orbitals have hybridised ☒B diagram shows sp² hybridisation as one 2s and two 2p orbitals have hybridised ☒C diagram shows sp hybridisation as one 2s and one 2p orbitals have hybridised ☒D none of the three 2p orbitals have hybridised in this diagram																									
13	B	92	<table><tr><td>Answer</td><td>A</td><td>B</td><td>C</td><td>D</td></tr><tr><td>Formula</td><td>C₂H₄</td><td>H₂O</td><td>O₂</td><td>N₂</td></tr><tr><td>Bonds</td><td>4x C-H 1x C=C</td><td>2x O-H</td><td>1x O=O</td><td>1x N≡N</td></tr><tr><td>Sigma Bonds</td><td>5</td><td>2</td><td>1</td><td>1</td></tr><tr><td>Pi Bonds</td><td>1</td><td>0</td><td>1</td><td>2</td></tr></table>	Answer	A	B	C	D	Formula	C ₂ H ₄	H ₂ O	O ₂	N ₂	Bonds	4x C-H 1x C=C	2x O-H	1x O=O	1x N≡N	Sigma Bonds	5	2	1	1	Pi Bonds	1	0	1	2
Answer	A	B	C	D																								
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14	A	54	Intermediate carbocation																									
15	D	50	A phenyl group is a benzene ring attached as a side-group. Benzene has a formula of C₆H₆ side group has a formula of -C₆H₅																									
16	C	74																										
17	B	57	ethoxy- side group on C₁ 3-methylbutane main chain																									
18	D	66	☒A H atom available on right side to rearrange C=O group to form -OH group ☒B H atom available on right side to rearrange C=O group to form -OH group ☒C H atom available on left side to rearrange C=O group to form -OH group ☒D no H atoms available on either side to rearrange C=O group to form -OH group																									

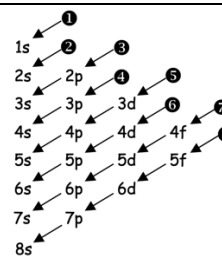
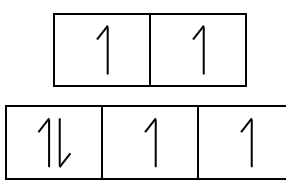
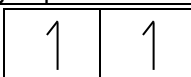
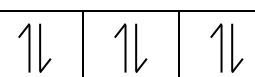
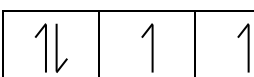
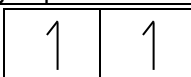
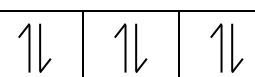
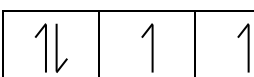
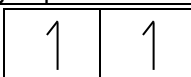
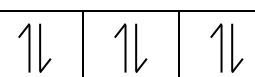
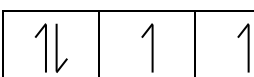
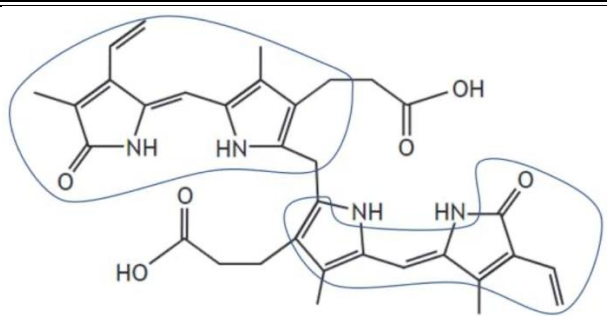
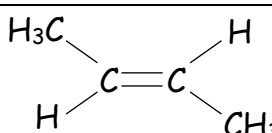
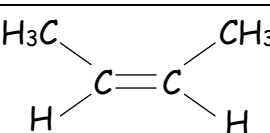
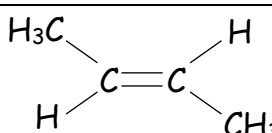
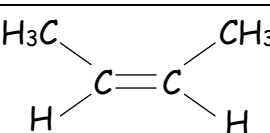
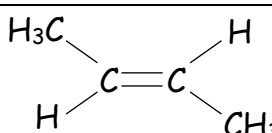
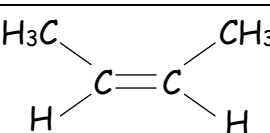
19	B	71	 A chiral carbon is a carbon with four different groups attached. This leads to non-superimposable mirror images in three dimensions 															
20	C	78	☒A Formula C_4H_4 ☒B Formula C_4H_6 ☒C Formula C_4H_8 fits empirical formula CH_2 ☒D Formula C_4H_{10} <table><tr><th>Element</th><th>C</th><th>H</th></tr><tr><td>Mass or %</td><td>0.12g</td><td>0.02g</td></tr><tr><td>No. of moles (divide % by gfm)</td><td>$\frac{0.12}{12}$ = 0.01</td><td>$\frac{0.02}{1}$ = 0.02</td></tr><tr><td>Mole ratio (divide through by smallest value)</td><td>$\frac{0.01}{0.01}$ = 1</td><td>$\frac{0.02}{0.01}$ = 2</td></tr><tr><td>Empirical Formula</td><td colspan="2">CH_2</td></tr></table>	Element	C	H	Mass or %	0.12g	0.02g	No. of moles (divide % by gfm)	$\frac{0.12}{12}$ = 0.01	$\frac{0.02}{1}$ = 0.02	Mole ratio (divide through by smallest value)	$\frac{0.01}{0.01}$ = 1	$\frac{0.02}{0.01}$ = 2	Empirical Formula	CH_2	
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Empirical Formula	CH_2																	
21	B	70	☒A $C_3H_8O_2$ has gfm= $(3 \times 12) + (8 \times 1) + (2 \times 16) = 36 + 8 + 32 = 76$ ☒B $C_3H_6O_2$ has gfm= $(3 \times 12) + (6 \times 1) + (2 \times 16) = 36 + 6 + 32 = 74$ ☒C C_3H_8O has gfm= $(3 \times 12) + (8 \times 1) + (1 \times 16) = 36 + 8 + 16 = 72$ ☒D C_3H_8O has gfm= $(3 \times 12) + (8 \times 1) + (1 \times 16) = 36 + 8 + 16 = 72$															
22	A	66	no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{6.7}{134} = 0.05\text{mol}$ sodium carbonate + malic acid $\longrightarrow$ sodium malate + water + carbon dioxide $\text{Na}_2\text{CO}_3 + \text{C}_4\text{H}_6\text{O}_5 \longrightarrow \text{Na}_2\text{C}_4\text{H}_4\text{O}_5 + \text{H}_2\text{O} + \text{CO}_2$ 1mol 1mol 0.05mol 0.05mol															
23	B	60	gfm $\text{CaCl}_2 \cdot 2\text{H}_2\text{O} = (1 \times 40.1) + (2 \times 35.5) + (4 \times 1) + (2 \times 16) = 40.1 + 71 + 4 + 32 = 147.1\text{g}$ gfm $\text{CaCl}_2 = (1 \times 40.1) + (2 \times 35.5) = 40.1 + 71 = 111.1\text{g}$ $\text{CaCl}_2 \cdot 2\text{H}_2\text{O} \longrightarrow \text{CaCl}_2 + 2\text{H}_2\text{O}$ 1mol 1mol 147.1g 111.1g 2.52g $111.1\text{g} \times \frac{2.52}{147.1}$ = 1.90g															
24	A	39	☒A Distillation will identify the boiling point of compound while purifying compound ☒B Recrystallisation will purify a compound but not help identify the compound ☒C Solvent Extraction will purify a compound but not help identify the compound ☒D Melting Point Determination will identify a compound but not purify compound															
25	B	38	☒A Impurities could move same distance as spot ☒B There is no reactant left (R) and only the product spot (S) is present ☒C More than one chemical could be present in the spot at (S) ☒D Some reactions at equilibrium neve reach completion															

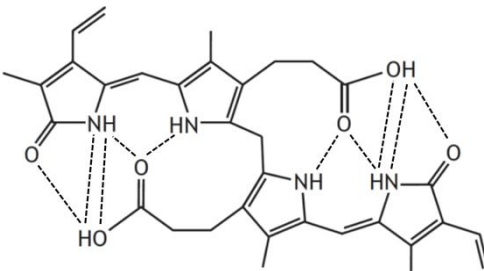
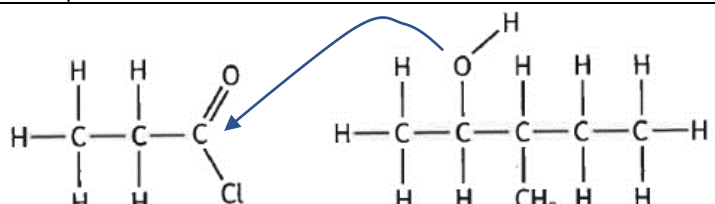
2022 Adv Higher Chemistry Marking Scheme

Long Qu	Answer	Reasoning																
1a(i)	∞ figure of 8 shape	<table><tr><th>Subshell Type</th><td>s</td><td>p</td><td>d</td></tr><tr><th>Angular Momentum Quantum number (l)</th><td>$l=0$</td><td>$l=1$</td><td>$l=2$</td></tr><tr><th>Shape(s)</th><td></td><td></td><td> or </td></tr></table>	Subshell Type	s	p	d	Angular Momentum Quantum number (l)	$l=0$	$l=1$	$l=2$	Shape(s)			 or 				
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1a(ii)	<table><tr><td>n</td><td>2</td></tr><tr><td>l</td><td>1</td></tr><tr><td>m_l</td><td>-1 or 0 or +1</td></tr><tr><td>m_s</td><td>$+\frac{1}{2}$ or $-\frac{1}{2}$</td></tr></table>	n	2	l	1	m_l	-1 or 0 or +1	m_s	$+\frac{1}{2}$ or $-\frac{1}{2}$	<table><tr><td>n</td><td>2p subshell is in 2nd shell and has a value for Principal quantum number (n) of n=2</td></tr><tr><td>l</td><td>p subshells have an angular momentum quantum number (l) of l =1</td></tr><tr><td>m_l</td><td>p subshells have a magnetic quantum number (m_l) of -l to +l $\therefore$ -1, 0, +1</td></tr><tr><td>m_s</td><td>All electrons either have a spin magnetic quantum number of either $+\frac{1}{2}$ or $-\frac{1}{2}$</td></tr></table>	n	2p subshell is in 2 nd shell and has a value for Principal quantum number (n) of n=2	l	p subshells have an angular momentum quantum number (l) of l =1	m_l	p subshells have a magnetic quantum number (m_l) of -l to +l $\therefore$ -1, 0, +1	m_s	All electrons either have a spin magnetic quantum number of either $+\frac{1}{2}$ or $-\frac{1}{2}$
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1b	Removing electron from oxygen creates a half-filled 2p orbital requires less energy as a half-filled p subshell is more stable. Or Removing an electron from nitrogen breaks a more stable half-filled 2p orbital which required more energy to break	<table><tr><th>Element</th><th>Nitrogen N</th><th>Oxygen O</th></tr><tr><td>Electronic configuration before electron removal</td><td>$1s^2 2s^2 2p^3$ </td><td>$1s^2 2s^2 2p^4$ </td></tr><tr><td>Electronic configuration after electron removal</td><td>$1s^2 2s^2 2p^2$ </td><td>$1s^2 2s^2 2p^3$ </td></tr><tr><td>Reasoning</td><td>Nitrogen atom has a half-filled 2p orbital and this is more stable electron configuration than the N⁺ ion with 2 electrons in the 2p orbital. More energy required to remove an electron from a half-filled subshell.</td><td>Oxygen atom has 4 electrons in 2p orbitals and losing an electron to become O⁺ ion creates the more stable half-filled subshell. Less energy is required to create a half-filled subshell by removing an electron.</td></tr></table>	Element	Nitrogen N	Oxygen O	Electronic configuration before electron removal	$1s^2 2s^2 2p^3$ 	$1s^2 2s^2 2p^4$ 	Electronic configuration after electron removal	$1s^2 2s^2 2p^2$ 	$1s^2 2s^2 2p^3$ 	Reasoning	Nitrogen atom has a half-filled 2p orbital and this is more stable electron configuration than the N ⁺ ion with 2 electrons in the 2p orbital. More energy required to remove an electron from a half-filled subshell.	Oxygen atom has 4 electrons in 2p orbitals and losing an electron to become O ⁺ ion creates the more stable half-filled subshell. Less energy is required to create a half-filled subshell by removing an electron.				
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1c	1.05×10^{-16}	$E = R \times Z^2 \times \left(1 - \frac{1}{n^2}\right)$ $= 2.18 \times 10^{-18} \times (8)^2 \times \left(1 - \frac{1}{2^2}\right)$ $= 2.18 \times 10^{-18} \times 64 \times (1 - 0.25)$ $= 2.18 \times 10^{-18} \times 64 \times (0.75)$ $= 1.0464 \times 10^{-16} \text{ J}$																
2a	Reaction not feasible at 298K as $\Delta G = +86.6 \text{ kJ mol}^{-1}$	<table><tr><td>$\begin{aligned} \Delta H^\circ &= \Sigma \Delta H_f^\circ(\text{products}) - \Sigma \Delta H_f^\circ(\text{reactants}) \\ &= (1 \times 90.3) - \left(\frac{1}{2} \times 0\right) + \left(\frac{1}{2} \times 0\right) \\ &= 90.3 - 0 \\ &= 90.3 \text{ kJ mol}^{-1} \end{aligned}$</td><td>$\begin{aligned} \Delta S^\circ &= \Sigma S^\circ(\text{products}) - \Sigma S^\circ(\text{reactants}) \\ &= (1 \times 211) - \left(\frac{1}{2} \times 192\right) + \left(\frac{1}{2} \times 205\right) \\ &= 211 - (96 + 102.5) \\ &= 211 - 198.5 \\ &= 12.5 \text{ J K}^{-1} \text{ mol}^{-1} \end{aligned}$</td></tr><tr><td colspan="2">$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = 90.3 - \left(298 \times \frac{12.5}{1000}\right) = 90.3 - (3.725) = +86.6 \text{ kJ mol}^{-1}$</td></tr></table>	$\begin{aligned} \Delta H^\circ &= \Sigma \Delta H_f^\circ(\text{products}) - \Sigma \Delta H_f^\circ(\text{reactants}) \\ &= (1 \times 90.3) - \left(\frac{1}{2} \times 0\right) + \left(\frac{1}{2} \times 0\right) \\ &= 90.3 - 0 \\ &= 90.3 \text{ kJ mol}^{-1} \end{aligned}$	$\begin{aligned} \Delta S^\circ &= \Sigma S^\circ(\text{products}) - \Sigma S^\circ(\text{reactants}) \\ &= (1 \times 211) - \left(\frac{1}{2} \times 192\right) + \left(\frac{1}{2} \times 205\right) \\ &= 211 - (96 + 102.5) \\ &= 211 - 198.5 \\ &= 12.5 \text{ J K}^{-1} \text{ mol}^{-1} \end{aligned}$	$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = 90.3 - \left(298 \times \frac{12.5}{1000}\right) = 90.3 - (3.725) = +86.6 \text{ kJ mol}^{-1}$													
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2b	600mg or 0.6g	5 litres per minute = 300 litres per hour 2ppm = 2mg per litre 1litre = 2mg 300 litres = $2\text{mg} \times \frac{300}{1}$ = 600 mg																
2c	<table><tr><td>+2</td></tr><tr><td>+4</td></tr><tr><td>+3</td></tr></table>	+2	+4	+3	<table><tr><td>NO</td><td>total oxidation state from O = $1 \times -2 = -2$</td><td>$\therefore$ Oxidation state of N = +2</td></tr><tr><td>NO₂</td><td>total oxidation state from O = $2 \times -2 = -4$</td><td>$\therefore$ Oxidation state of N = +4</td></tr><tr><td>N₂O₃</td><td>total oxidation state from O = $3 \times -2 = -6$</td><td>$\therefore$ Oxidation State of N = $+\frac{6}{2} = +3$</td></tr></table>	NO	total oxidation state from O = $1 \times -2 = -2$	$\therefore$ Oxidation state of N = +2	NO ₂	total oxidation state from O = $2 \times -2 = -4$	$\therefore$ Oxidation state of N = +4	N ₂ O ₃	total oxidation state from O = $3 \times -2 = -6$	$\therefore$ Oxidation State of N = $+\frac{6}{2} = +3$				
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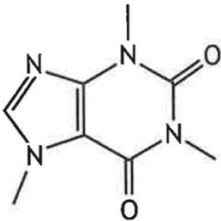
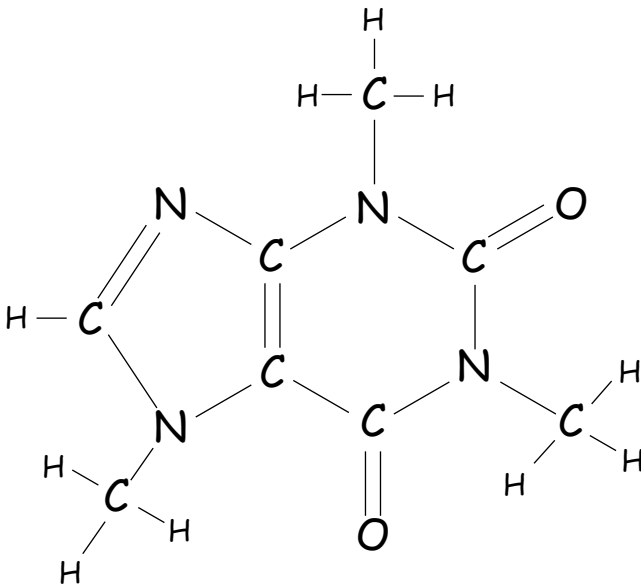
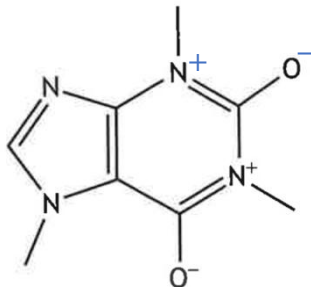
2d(i)A	Rate = k [NO] ² [H ₂]	<table><tr><th>no. of mol in slow step</th><th>Order of Reactant</th><th>Effect on Reactant</th><th>Rate Equation</th></tr><tr><td>2mol NO</td><td>2nd order</td><td>[NO]²</td><td rowspan="2">Rate = k [NO]² [H₂]</td></tr><tr><td>1mol H₂</td><td>1st order</td><td>[H₂]</td></tr></table>	no. of mol in slow step	Order of Reactant	Effect on Reactant	Rate Equation	2mol NO	2 nd order	[NO] ²	Rate = k [NO] ² [H ₂]	1mol H ₂	1 st order	[H ₂]
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1mol H ₂	1 st order	[H ₂]											
2d(i)B	9.94×10 ⁻³ or 0.0099	$[\text{NO}]^2 = \frac{\text{Rate}}{K[\text{H}_2]} = \frac{0.0040}{2.7 \times 10^3 \times 0.015} = 9.87 \times 10^{-5}$ $[\text{NO}] = 9.94 \times 10^{-3} \text{ mol l}^{-1}$											
2d(ii)	Equation showing:	2NO + 2H ₂ → N ₂ + 2H ₂ O											
3a	acid partially dissociates into ions	Strong acids are acids which fully dissociates into ions Weak acids are acids with partially dissociate into ions.											
3b(i)	1.298	$\text{pH} = \frac{1}{2}\text{pK}_a - \frac{1}{2}\log_{10}c$ $\text{pH} = \frac{1}{2}(3.17) - \frac{1}{2}\log_{10}(3.75)$ $\text{pH} = 1.585 - \frac{1}{2}(0.574)$ $\text{pH} = 1.585 - 0.287$ $\text{pH} = 1.298$											
3b(ii)	7.5%	gfm HF = (1×1)+(1×19) = 1+19 = 20g 3.75mol of HF in 1litre of solution ∴ mass = no. of mol × gfm = 3.75 × 20 = 75g 1 litre of HF solution would have approx. mass of 1000g $\frac{75\text{g}}{1000\text{g}} \times 100 = 7.5\%$											
3b(iii)	tetrahedral	<table><tr><td>2 electron pairs 2 bonding + 0 lone pairs  Linear</td><td>3 electron pairs 3 bonding + 0 lone pairs  Trigonal Planar</td><td>4 electron pairs 4 bonding + 0 lone pairs  Tetrahedral</td><td>4 electron pairs 3 bonding + 1 lone pair  Trigonal Pyramidal</td><td>4 electron pairs 2 bonding + 2 lone pairs  Angular</td><td>5 electron pairs 5 bonding + 0 lone pairs  Trigonal Bipyramidal</td><td>5 electron pairs 3 bonding + 2 lone pairs  T-Shaped</td><td>6 electron pairs 6 bonding + 0 lone pairs  Octahedral</td><td>6 electron pairs 4 bonding + 2 lone pairs  Square Planar</td></tr></table> $\text{Electron pairs} = \frac{\text{No of outer electrons on central atom} + \text{no. of bonds} - \text{charge}}{2}$ $= \frac{4 + 4 - 0}{2} = \frac{8}{2} = 4 \text{ pairs of electrons (4 bonding pairs)}$	2 electron pairs 2 bonding + 0 lone pairs  Linear	3 electron pairs 3 bonding + 0 lone pairs  Trigonal Planar	4 electron pairs 4 bonding + 0 lone pairs  Tetrahedral	4 electron pairs 3 bonding + 1 lone pair  Trigonal Pyramidal	4 electron pairs 2 bonding + 2 lone pairs  Angular	5 electron pairs 5 bonding + 0 lone pairs  Trigonal Bipyramidal	5 electron pairs 3 bonding + 2 lone pairs  T-Shaped	6 electron pairs 6 bonding + 0 lone pairs  Octahedral	6 electron pairs 4 bonding + 2 lone pairs  Square Planar		
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4a(i)A	Heating under reflux	Heating under reflux allows heating of a chemical reaction mixture over an extended period of time without volatile substances escaping a round-bottomed flask is used containing anti-bumping granules to reduce bumping of reactants in the flask											
4a(i)B	Anti-bumping granules	- flask is fitted with a condenser to allow condensation of reaction mixture vapour and prevents escape of volatile chemicals - flask is heated using an appropriate heat source e.g. heating mantle it there are flammable reactant/products involved.											
4a(ii)	Answer to include:	 few ions											

4a(iii)A	Buchner funnel with vacuum filtration	Vacuum filtration: filtration under reduced pressure faster means of separating a precipitate from a filtrate Büchner, Hirsch or sintered glass funnel can be used.																		
4a(iii)B	To purify the hippuric acid	Recrystallisation: purify an impure solid with: - dissolving an impure solid gently in a minimum volume of a hot solvent - hot filtration of resulting mixture to remove any insoluble impurities - cooling the filtrate slowly to allow crystals of the pure compound to form, leaving soluble impurities dissolved in the solvent - filtering, washing and drying the pure crystals solvent for recrystallisation is chosen so that the compound being purified is completely soluble at high temperatures and only sparingly soluble at lower temperatures.																		
4b(i)	Different atoms types in bonds and different strengths of bonds decides which wavelengths/ wavenumbers of infrared are absorbed.	When infrared radiation is absorbed by organic compounds, bonds within the molecule vibrate/stretch/bend - wavelengths of infrared radiation absorbed depend on the type of atoms that make up the bond and the strength of the bond - infrared radiation is passed through a sample of the organic compound and then into a detector that measures the intensity of the transmitted radiation at different wavelengths. The absorbance of infrared radiation is measured in wavenumbers, measured in cm^{-1}																		
4b(ii)	3340 to 3400	Information found in data booklet.																		
4c(i)	6	<table><tr><th>^1H Environment</th><th>Grouping</th></tr><tr><td>①</td><td>ArH (9.0-6.9)</td></tr><tr><td>②</td><td>Amide link (9.4-5.0)</td></tr><tr><td>③</td><td>-CH₂- (?)</td></tr><tr><td>④</td><td>R-CH₂-COOH (2.7-2.0)</td></tr></table> 	^1H Environment	Grouping	①	ArH (9.0-6.9)	②	Amide link (9.4-5.0)	③	-CH ₂ - (?)	④	R-CH ₂ -COOH (2.7-2.0)								
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4c(iii)	<u>1st mark:</u> Hydrogen nuclei flip to align with magnetic field <u>2nd mark:</u> Different energy for different hydrogen environments	^1H nuclei behave like tiny magnets and in a strong magnetic field some align with the field (lower energy), whilst the rest align against it (higher energy). - Absorption of radiation in the radio frequency region of the electromagnetic spectrum causes the ^1H nuclei to 'flip' from the lower to the higher energy alignment. - As they fall back from the higher to the lower energy alignment the emitted radiation is detected and plotted on a spectrum. - In a ^1H NMR spectrum the chemical shift, δ, (peak position) is related to the environment of the ^1H atom and is measured in parts per million (ppm). The area under the peak is related to the number of ^1H atoms in that environment and is often given by an integration curve on a spectrum. The height of an integration curve is proportional to the number of ^1H atoms in that environment, and so a ratio of ^1H atoms in each environment can be determined.																		
4d	Open Ended Question: <table><tr><th colspan="4">Candidate Marks</th></tr><tr><td>0 marks</td><td>1 mark</td><td>2 mark</td><td>3 marks</td></tr><tr><td>43%</td><td>41%</td><td>13%</td><td>4%</td></tr></table>	Candidate Marks				0 marks	1 mark	2 mark	3 marks	43%	41%	13%	4%	<table><tr><th>3 mark answer</th><th>2 mark answer</th><th>1 mark answer</th></tr><tr><td>Demonstrates a <u>good</u> understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.</td><td>Demonstrates a <u>reasonable</u> understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.</td><td>Demonstrates a <u>limited</u> understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.</td></tr></table>	3 mark answer	2 mark answer	1 mark answer	Demonstrates a <u>good</u> understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a <u>reasonable</u> understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited</u> understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.
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5a	Repulsion from electrons/lone pair	Ligands may be negative ions or molecules with non-bonding pairs of electrons that they donate to the central metal atom or ion, forming dative covalent bonds. Electrons in approaching ligands cause the splitting of d orbitals into higher and lower energies as the electrons along the axes to be repelled. - ligands that cause a large difference in energy between subsets of d orbitals are strong field ligands. - weak field ligands cause a small energy difference. - spectrochemical series is a list of ligands placed in order of their ability to split d orbitals																																				
5b(i)	Orbitals fill orbitals in order of increasing energy	<u>Aufbau Principle:</u> Electrons fill up in order of increasing energy: 1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p 6s 4f 5d 6p 7s 5f 6d 7p 8s 																																				
5b(ii)A	Both conclusions required:	<table><tr><td>Conclusion 1 Higher charge of Cobalt in complex gives higher energy required to pair two electrons with opposite spin (P)</td><td>Complex Ion</td><td>$[\text{Co}(\text{OH}_2)_6]^{2+}$</td><td>$[\text{CoF}_6]^{3-}$</td><td>$[\text{Co}(\text{OH}_2)_6]^{3+}$</td><td>$[\text{Co}(\text{CN})_6]^{3-}$</td></tr><tr><td>Conclusion 2: Higher spin state has lower difference in energy between subsets of d orbitals (Δ)</td><td>Charge on Co</td><td>Co^{2+}</td><td>Co^{3+}</td><td>Co^{3+}</td><td>Co^{3+}</td></tr><tr><td></td><td>P (kJ mol^{-1})</td><td>250</td><td>283</td><td>283</td><td>283</td></tr></table> <table><tr><td>Δ (kJ mol^{-1})</td><td>111</td><td>156</td><td>323</td><td>416</td></tr><tr><td>Spin State</td><td>High</td><td>High</td><td>Low</td><td>Low</td></tr></table>	Conclusion 1 Higher charge of Cobalt in complex gives higher energy required to pair two electrons with opposite spin (P)	Complex Ion	$[\text{Co}(\text{OH}_2)_6]^{2+}$	$[\text{CoF}_6]^{3-}$	$[\text{Co}(\text{OH}_2)_6]^{3+}$	$[\text{Co}(\text{CN})_6]^{3-}$	Conclusion 2: Higher spin state has lower difference in energy between subsets of d orbitals (Δ)	Charge on Co	Co^{2+}	Co^{3+}	Co^{3+}	Co^{3+}		P (kJ mol^{-1})	250	283	283	283	Δ (kJ mol^{-1})	111	156	323	416	Spin State	High	High	Low	Low								
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Δ (kJ mol^{-1})	111	156	323	416																																		
Spin State	High	High	Low	Low																																		
5b(ii)B	Hexafluoridocobaltate (III)	Hexafluoridocobaltate(III) = $[\text{CoF}_6]^{3-}$ <table><tr><td>no. of ligands</td><td>6</td><td>F⁻ ligand</td><td>metal name</td><td>cobalt</td><td>negative charge on complex</td><td>-3</td><td>charge on metal ion</td><td>+3</td></tr></table> <table><tr><td colspan="2">Neutral ligands include:</td><td colspan="2">Negative Ligands include:</td><td>Central Ion:</td><td>Charge:</td></tr><tr><td>Ligand</td><td>Name</td><td>Ligand</td><td>Name</td><td>Positive Complex: metals keep their name</td><td rowspan="4">Charge of central ion is converted into roman numerals and put in brackets</td></tr><tr><td>H₂O</td><td>aqua</td><td>Chloride Cl⁻</td><td>chlorido</td><td>Negative Complex: Metals end in ATE</td></tr><tr><td>NH₃</td><td>ammine</td><td>Cyanide CN⁻</td><td>cyanido</td><td>Metals end in ATE</td></tr><tr><td>CO</td><td>carbonyl</td><td>Nitrite NO₂⁻</td><td>nitrito</td><td>e.g. Cuprate, Ferrate, Cobaltate</td></tr></table>	no. of ligands	6	F ⁻ ligand	metal name	cobalt	negative charge on complex	-3	charge on metal ion	+3	Neutral ligands include:		Negative Ligands include:		Central Ion:	Charge:	Ligand	Name	Ligand	Name	Positive Complex: metals keep their name	Charge of central ion is converted into roman numerals and put in brackets	H ₂ O	aqua	Chloride Cl ⁻	chlorido	Negative Complex: Metals end in ATE	NH ₃	ammine	Cyanide CN ⁻	cyanido	Metals end in ATE	CO	carbonyl	Nitrite NO ₂ ⁻	nitrito	e.g. Cuprate, Ferrate, Cobaltate
no. of ligands	6	F ⁻ ligand	metal name	cobalt	negative charge on complex	-3	charge on metal ion	+3																														
Neutral ligands include:		Negative Ligands include:		Central Ion:	Charge:																																	
Ligand	Name	Ligand	Name	Positive Complex: metals keep their name	Charge of central ion is converted into roman numerals and put in brackets																																	
H ₂ O	aqua	Chloride Cl ⁻	chlorido	Negative Complex: Metals end in ATE																																		
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CO	carbonyl	Nitrite NO ₂ ⁻	nitrito	e.g. Cuprate, Ferrate, Cobaltate																																		
5b(ii)C		<table><tr><td>Electronic Configuration</td><td>Low Spin state of Co^{3+}</td><td>High Spin State of Co^{3+}</td></tr><tr><td>Element Cobalt Configuration: 1s² 2s² 2p⁶ 3s² 3p⁶ 3d⁷ 4s²</td><td></td><td></td></tr><tr><td>Cobalt Co^{3+} Ion Configuration: 1s² 2s² 2p⁶ 3s² 3p⁶ 3d⁶</td><td></td><td></td></tr></table>	Electronic Configuration	Low Spin state of Co^{3+}	High Spin State of Co^{3+}	Element Cobalt Configuration: 1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 3d ⁷ 4s ²			Cobalt Co^{3+} Ion Configuration: 1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 3d ⁶																													
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Cobalt Co^{3+} Ion Configuration: 1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 3d ⁶																																						
6a(i)	Circled region of alternating C=C double bonds and C-C single bond.																																					
6a(ii)	Antibonding molecular orbitals are higher in energy	Electrons fill bonding molecular orbitals, leaving higher energy antibonding molecular orbitals unfilled.																																				
6b(i)	Answer to include:	C=C double bonds do not rotate around their axis to geometric isomers exist depending on the position of side groups in relation to C=C double bond. <table><tr><td>-CH₃ methyl groups are on opposite sides of the C=C double bond</td><td>-CH₃ methyl groups are on same side of the C=C double bond</td></tr><tr><td> trans-but-2-ene </td><td> cis-but-2-ene </td></tr></table>	-CH ₃ methyl groups are on opposite sides of the C=C double bond	-CH ₃ methyl groups are on same side of the C=C double bond	 trans-but-2-ene	 cis-but-2-ene																																
-CH ₃ methyl groups are on opposite sides of the C=C double bond	-CH ₃ methyl groups are on same side of the C=C double bond																																					
 trans-but-2-ene	 cis-but-2-ene																																					

6b(ii)	Dotted lines on diagram showing:	Dotted lines must be between: H of an N-H or O-H bond and O of a O-H bond or N of N-H bond			
6c(i)	Blue light provides enough energy to break bonds in bilirubin	Blue light has shorter wavelength than red light and has more energy.			
6c(ii)	257.5	$E = \frac{L \times h \times c}{\lambda} = \frac{6.02 \times 10^{23} \text{ mol}^{-1} \times 6.63 \times 10^{-34} \text{ J s} \times 3 \times 10^8 \text{ m s}^{-1}}{465 \times 10^{-9} \text{ m}}$ $= 257501 \text{ J mol}^{-1}$ $= 257.5 \text{ kJ mol}^{-1}$			
7a(i)	Answer to include:	Place empty weighing bottle or weighing boat on balance. Record the mass of the empty item on balance. Add substance to boat/bottle on balance (careful not to spill any onto the top of balance). Record mass and subtract the empty mass to get mass of substance.			
7a(ii)	0.0204	Absorbance = 0.42 $\therefore [\text{MnO}_4^-] = 2.1 \times 10^{-4} \text{ mol l}^{-1}$ no. of mol = volume $\times$ concentration = $0.1 \text{ litres} \times 2.1 \times 10^{-4} \text{ mol l}^{-1} = 0.000021 \text{ mol}$ gfm = 54.9g mass = no. of mol $\times$ gfm = $0.000021 \text{ mol} \times 54.9 \text{ g mol}^{-1} = 0.00115 \text{ g}$ $\% \text{ mass} = \frac{0.00115}{5.66} \times 100 = 0.0204\%$			
7b(i)	one from:	high state of purity	be stable when solid and in solution	be soluble	reasonably high GFM
7b(ii)	0.358	no. of mol $\text{Cr}_2\text{O}_7^{2-}$ = volume $\times$ concentration = $0.0214 \text{ litres} \times 0.005 \text{ mol l}^{-1} = 0.000107 \text{ mol}$ $6\text{Fe}^{2+} + \text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ \longrightarrow 6\text{Fe}^{3+} + 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$ 6mol 1mol 0.000642mol 0.000107mol 25cm ³ Fe ²⁺ solution = 0.000642mol 250cm ³ Fe ²⁺ solution = 0.00642mol gfm Fe = 55.8g mass = no. of mol $\times$ gfm = $0.00642 \text{ mol} \times 55.8 \text{ g mol}^{-1} = 0.358 \text{ g}$			
7b(iii)	Answer to contain:	1 mark Green wavelengths absorbed turning HOMO into LUMO		1 mark Green wavelengths absorbed $\therefore$ red + blue wavelengths transmitted and purple light emitted	
8a	H ₃ PO ₃ (any order of elements)	$3\text{CH}_3\text{COOH} + \text{PCl}_3 \longrightarrow 3\text{CH}_3\text{COCl} + \text{H}_3\text{PO}_3$			
8b(i)	Negatively charged ions/neutral molecules that are electron rich	Nucleophiles are negatively charged ions or neutral molecules which are electron rich - attracted towards atoms bearing a partial δ^+ or full positive charge - capable of donating an electron pair to form a new covalent bond Examples: Cl ⁻ , Br ⁻ , OH ⁻ , CN ⁻ , NH ₃ and H ₂ O Electrophiles are positively charged ions or neutral molecules that are electron deficient - attracted towards atoms bearing a partial δ^- or full negative charge - capable of accepting an electron pair to form a new covalent bond Examples: H ⁺ , NO ₂ ⁺ and SO ₃			
8b(ii)	Curly arrow from O on alcohol going to C of acid chloride group				

8b(iii)	Diagram showing:	H H O H CH₃ H H H C C C O C C C C H H H CH₃ H H H												
8b(iv)	HCl	Condensation reactions join two molecules together and remove a small molecule as they join. Water is the most likely small molecule removed but hydrogen chloride HCl is removed in the condensation of alcohols and acid chloride to form esters.												
8b(v)	Faster reaction or catalyst not required	As the equilibrium lies more to the product side in this reaction then more ester is produced at equilibrium												
8c(i)	Secondary	Primary Amine	Secondary Amine	Tertiary Amine										
		H N C₃H₇ H	H₃C N C₂H₅ H	H₃C N CH₃ CH₃										
		1 Carbon attached to the Nitrogen	2 Carbons attached to the Nitrogen	3 Carbons attached to the Nitrogen										
8c(ii)	amide	Amide links have a carbonyl group attached to a nitrogen which can have one or two carbon groups attached to the nitrogen.	O H - C - N -	O CH₃ - C - N -										
8d(i)	Electrophilic substitution	AlCl ₃ catalyst polarises the C-Cl bond in benzoyl chloride and the carbon end joins the benzene ring by electrophilic substitution.												
8d(ii)	65.2%	Benzoyl chloride no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{21.8}{140.5} = 0.155\text{mol}$		Benzophenone no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{18.4}{182} = 0.101\text{mol}$ (actual)										
		benzoyl chloride + benzene $\longrightarrow$ benzophenone + hydrogen chloride 1mol 0.155mol 1mol 0.155mol (theoretical) $\% \text{ Yield} = \frac{\text{Actual}}{\text{Theoretical}} \times 100 = \frac{0.101\text{mol}}{0.155\text{mol}} \times 100 = 65.2\%$												
9	Open Ended Question:	3 mark answer	2 mark answer	1 mark answer										
		Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.										
		<table><tr><th colspan="4">Candidate Marks</th></tr><tr><td>0 marks</td><td>1 mark</td><td>2 marks</td><td>3 marks</td></tr><tr><td>23%</td><td>37%</td><td>29%</td><td>11%</td></tr></table>			Candidate Marks				0 marks	1 mark	2 marks	3 marks	23%	37%
Candidate Marks														
0 marks	1 mark	2 marks	3 marks											
23%	37%	29%	11%											
10a	Antagonist binds to receptor preventing the natural substrate from binding and stops natural response.	agonist	An agonist mimics the natural compound and binds to the receptor molecules to produce a response similar to the natural active compound.											
		antagonist	An antagonist prevents the natural compound from binding to the receptor, and so blocks the natural response from occurring.											
10b(i)	5cm ³	A one in one hundred dilution requires a 5cm ³ of eucalyptol transferred by pipette into a 500cm ³ standard/volumetric flask and the filled up to the mark with deionised water.												
10b(ii)	0.0598	mass = density x Volume = 0.921g cm⁻³ x 5cm³ = 4.605g no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{4.605\text{ g}}{154\text{ g mol}^{-1}} = 0.0299\text{mol}$ concentration = $\frac{\text{no. of mol}}{\text{Volume}} = \frac{0.0299\text{ mol}}{0.500\text{ litres}} = 0.0598\text{mol l}^{-1}$												

10b(iii)	6	no. of mol = Volume $\times$ concentration = 1 litre $\times$ $9.97 \times 10^{-24} \text{ mol l}^{-1}$ = $9.97 \times 10^{-24} \text{ mol}$ $1 \text{ mol} = 6.02 \times 10^{23} \text{ molecules}$ $9.97 \times 10^{-24} \text{ mol} = 6.02 \times 10^{23} \text{ molecules} \times \frac{9.97 \times 10^{-24}}{1}$ $= 6.00 \text{ molecules}$	
11a(i)	$\text{C}_8\text{H}_{10}\text{N}_4\text{O}_2$ (any order of elements)		
		Skeletal Structure	Full Structural Formula
11a(ii)	Molecules must have similar shape to bind to same receptor protein	Pharmacologically active molecules must have a similar shape to fit the same receptor molecule. This common shape is called the pharmacophore. Agonists fit the receptor and produce the same biological response as the natural substrate. Antagonist fit the receptor but do not cause the biological response inside the cell.	
11b(i)A	<u>1st mark:</u> separating funnel <u>2nd mark:</u> Shake/mix Leave to separate Run off lower layer	The steps of recrystallisation to purify an impure solid include: - dissolving an impure solid gently in a minimum volume of a hot solvent - hot filtration of the resulting mixture to remove any insoluble impurities - cooling the filtrate slowly to allow crystals of the pure compound to form, leaving soluble impurities dissolved in the solvent - filtering, washing and drying the pure crystals The solvent used for recrystallisation is chosen so that the compound being purified is completely soluble at high temperatures and only sparingly soluble at lower temperatures. The solvent used should be: - immiscible with the liquid mixture or solution (usually water) - one in which the solute is more soluble in than the liquid mixture or solution (usually water) - volatile to allow the solute to be obtained by evaporation of the solvent - unreactive with the solute	
11b(i)B	4.61	$K = \frac{[\text{caffeine}]_{\text{dichloromethane}}}{[\text{caffeine}]_{\text{water}}} = \frac{23.5/60}{8.5/100} = \frac{0.392}{0.085} = 4.61$	
11b(ii)	Do extraction in three separate volumes of 20cm^3	The quantity of caffeine extracted is greater if a number of extractions using smaller volumes of solvent are carried out rather than a single extraction using a large volume of solvent.	
11c	Diagram completed as shown:		

2022 Advanced Higher Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	B	3	66	5	26	0	0
2	D	12	7	21	60	0	0
3	D	13	19	6	61	1	0
4	A	83	9	2	5	0	0
5	D	6	28	23	41	2	0
6	C	38	12	45	4	1	0
7	A	54	26	4	15	0	0
8	D	10	8	19	62	0	0
9	B	5	83	5	7	0	0
10	D	8	7	9	77	0	0
11	C	15	8	65	11	0	0
12	C	20	11	54	14	1	0
13	B	4	92	2	2	0	0
14	A	54	13	20	13	0	0
15	D	22	17	10	50	1	0
16	C	7	9	74	9	0	0
17	B	2	57	6	35	0	0
18	D	5	10	17	66	1	0
19	B	5	71	10	13	0	0
20	C	11	6	78	4	1	0
21	B	17	70	6	6	1	0
22	A	66	9	15	8	1	0
23	B	14	60	14	10	1	0
24	A	39	28	10	23	0	0
25	B	29	38	9	24	1	0

2022 Advanced Higher Chemistry Section 2 Marks Data Analysis

2022 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1ai	1	0.94	94%	27	0%
Q1aai	1	0.81	81%	35	0%
Q1b	1	0.52	52%	34	1%
Q1c	1	0.64	64%	24	1%
Q2a	3	2.32	77%	55	2%
Q2b	2	1.19	60%	53	2%
Q2c	1	0.84	84%	42	1%
Q2diA	1	0.61	61%	47	2%
Q2diB	1	0.60	60%	48	4%
Q2dii	1	0.70	70%	43	6%
Q3a	1	0.57	57%	36	2%
Q3bi	2	1.80	90%	39	1%
Q3bii	2	0.49	24%	42	10%
Q3biii	1	0.84	84%	29	1%
Q4aiA	1	0.82	82%	48	2%
Q4aiB	1	0.57	57%	43	3%
Q4aii	1	0.16	16%	37	11%
Q4aiiiA	1	0.74	74%	48	7%
Q4aiiiB	1	0.87	87%	37	3%
Q4bi	1	0.50	50%	32	4%
Q4bii	1	0.77	77%	25	1%
Q4ci	1	0.08	8%	7	1%
Q4cii	1	0.51	51%	46	11%
Q4ciii	2	0.25	12%	37	22%
Q4d	3	0.77	26%	54	8%
Q5a	1	0.17	17%	30	7%
Q5bi	1	0.61	61%	35	1%
Q5biiA	2	1.30	65%	30	3%
Q5biiB	1	0.32	32%	35	1%
Q5biiC	1	0.31	31%	36	1%
Q6ai	1	0.09	9%	15	18%
Q6aii	1	0.42	42%	42	8%
Q6bi	2	0.76	38%	54	5%
Q6bii	1	0.45	45%	27	20%
Q6ci	1	0.60	60%	26	2%
Q6cii	2	1.69	85%	37	1%
Q7ai	2	1.12	56%	39	9%
Q7aii	2	1.35	68%	56	4%
Q7bi	1	0.80	80%	44	3%
Q7bii	3	2.32	77%	61	4%
Q7biii	2	0.99	50%	52	4%
Q8a	1	0.30	30%	34	2%
Q8bi	1	0.35	35%	31	2%
Q8bii	1	0.13	13%	29	3%
Q8biii	1	0.40	40%	43	5%
Q8biv	1	0.29	29%	19	4%
Q8bv	1	0.49	49%	45	8%
Q8ci	1	0.83	83%	49	5%
Q8cii	1	0.26	26%	34	4%
Q8di	1	0.69	69%	26	1%
Q8dii	2	1.66	83%	46	2%
Q9	3	1.27	42%	65	13%
Q10a	1	0.34	34%	29	2%
Q10bi	1	0.85	85%	28	3%
Q10bii	2	0.89	44%	40	9%
Q10biii	1	0.49	49%	43	12%
Q11ai	1	0.66	66%	35	2%
Q11aii	1	0.54	54%	26	3%
Q11biA	2	0.47	24%	49	17%
Q11biB	3	1.02	34%	55	7%
Q11bii	1	0.14	14%	33	16%
Q11c	1	0.80	80%	25	5%
Paper	85	45.11	53%		

2022 Advanced Higher Chemistry Section 2 Frequency of Marks Analysis

2022 Item	0	1	2	3
Q1ai	6	94		
Q1aii	19	81		
Q1b	48	52		
Q1c	36	64		
Q2a	4	11	36	50
Q2b	28	25	47	
Q2c	16	84		
Q2diA	39	61		
Q2diB	40	60		
Q2dii	30	70		
Q3a	43	57		
Q3bi	7	6	87	
Q3bii	66	20	15	
Q3biii	16	84		
Q4aiA	18	82		
Q4aiB	43	57		
Q4aii	84	16		
Q4aiiiA	26	74		
Q4aiiiB	13	87		
Q4bi	50	50		
Q4bii	23	77		
Q4ci	92	8		
Q4cii	49	51		
Q4ciii	82	12	6	
Q4d	43	41	13	4
Q5a	83	17		
Q5bi	39	61		
Q5biiA	13	43	43	
Q5biiB	68	32		
Q5biiC	69	31		
Q6ai	91	9		
Q6aii	58	42		
Q6bi	36	51	13	
Q6bii	55	45		
Q6ci	40	60		
Q6cii	3	26	72	
Q7ai	36	16	48	
Q7aii	10	44	46	
Q7bi	20	80		
Q7bii	9	5	29	56
Q7biii	32	36	32	
Q8a	70	30		
Q8bi	65	35		
Q8bii	87	13		
Q8biii	60	40		
Q8biv	71	29		
Q8bv	51	49		
Q8ci	17	83		
Q8cii	74	26		
Q8di	31	69		
Q8dii	11	11	77	
Q9	23	37	29	11
Q10a	66	34		
Q10bi	15	85		
Q10bii	37	36	26	
Q10biii	51	49		
Q11ai	34	66		
Q11aii	46	54		
Q11biA	62	28	10	
Q11biB	39	29	24	9
Q11bii	86	14		
Q11c	20	80		